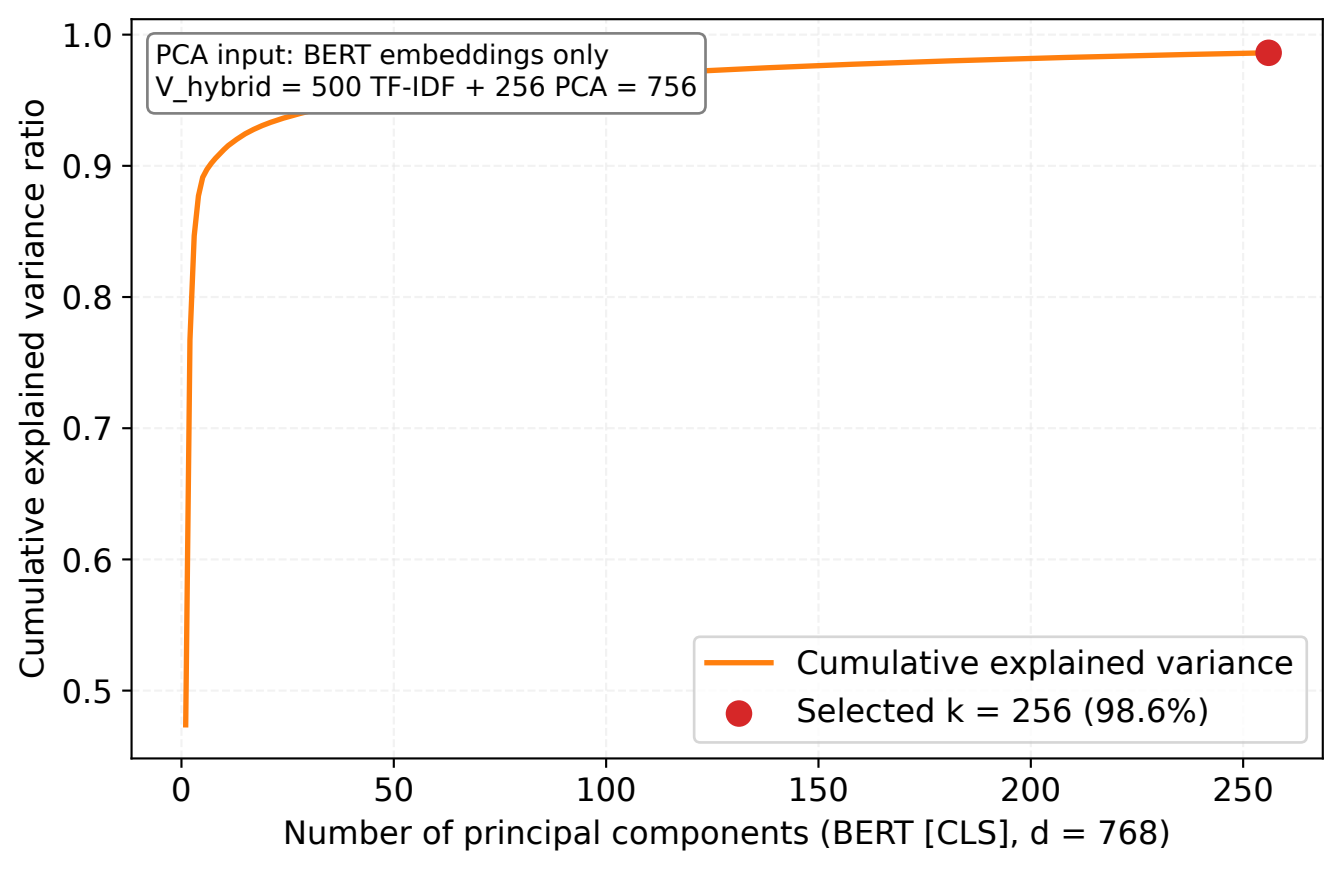
PCA input: BERT embeddings only
V_hybrid = 500 TF-IDF + 256 PCA = 756
Cumulative explained variance ratio
1.0
0.9
0.8
0.7
0.6
0.5
0
50
100
150
200
250
Number of principal components (BERT [CLS], d = 768)
Cumulative explained variance
Selected k = 256 (98.6%)